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14 AUGUST 2026

DESCRIBE INFORMATION SYSTEMS DEPARTMENTS IN BUSINESS ORGANISATIONS

US 14926

SECTION A : FORMATIVE

Question 1

1. CIO - Chief Information Officer

Purpose: Overall strategic leader of IT.

Tasks: Aligns IT with business goals, budgeting, liaises with executive management, sets IT policy.

2. IT Manager / Information Systems Manager

Purpose: Manages day-to-day operations of the IS department.

Tasks: Supervises teams, ensures systems run smoothly, manages staff and projects.

3. Systems Analyst

Purpose: Bridge between business and IT.

Tasks: Gathers requirements, analyzes business processes, writes system specifications.

4. Systems Designer

Purpose: Designs the technical solution.

Tasks: Creates system architecture, database design, UI/UX design, technical blueprints.

5. Systems Programmer / Developer

Purpose: Builds the software.

Tasks: Codes, tests, debugs and maintains applications based on designer specs.

6. Network Administrator

Purpose: Manages company networks.

Tasks: Installs, configures and secures LAN, WAN, firewalls for the bank.

7. Database Administrator DBA

Purpose: Manages data.

Tasks: Ensures data security, backup, performance and integrity. Critical for POPIA in banks.

8. Help Desk / Technical Support

Purpose: User support.

Tasks: Resolves staff IT issues, hardware/software troubleshooting.

9. QA Tester

Purpose: Quality assurance.

Tasks: Tests systems for defects before release.

10. Information Security Officer

Purpose: Protects bank systems and data.

Tasks: Manages cybersecurity, access control, compliance with POPIA and FICA.

Question 2

Compare the development tasks of the systems analyst, systems programmer and systems designer.*

Role Main Focus Key Development Tasks

Systems Analyst: "What the system must do"

1. Meet with users to gather requirements

2. Analyze current business processes 3. Document functional requirements 4. Evaluate feasibility

Systems Designer: "How the system will work"

1. Design system architecture and databases

2. Design user interfaces and reports

3. Create technical specifications for programmers 4. Plan integration between modules

Systems Programmer :"Build the system"

1. Write code in programming languages
2. . Test and debug modules
3. Implement designer's technical specs

4.Maintain and update code

Summary: Analyst defines requirements, Designer plans the solution, Programmer builds it.

Question 3

a) Organizational Chart

For a large bank, use a Centralized model for better security and control.

CIO - Chief Information Officer

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├– IS Manager: Systems Development

| ├– Senior Systems Analyst x2

| ├– Systems Designer x2

| ├– Senior Programmer x3

| └– QA Tester x2

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├– IS Manager: Infrastructure & Networking

| ├– Network Administrator x3

| ├– Systems Engineer x2

| ├– Server Administrator x2

| └– IT Support Technician x4

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├– IS Manager: Database & Security

| ├– Database Administrator DBA x3

| ├– Information Security Officer x2

| ├– Cyber Security Analyst x2

| └– Backup Administrator x1

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└– IS Manager: User Support & Training

  ├– Help Desk Supervisor x1

  ├– Help Desk Agent x6

  └– IT Trainer x2

b. i. Centralized vs Decentralized

Decision: Centralized Model*

Reason: Banks need strict control, standard security, and centralized data due to POPIA, FICA, and financial regulations. All IT decisions come from head office.

Top Post: CIO - Chief Information Officer

b. ii. Teams/Units - First 2 layers

1. Systems Development- Headed by IS Manager: Systems Development

2. Infrastructure & Networking - Headed by IS Manager: Infrastructure

3. Database & Security- Headed by IS Manager: Database & Security

4. User Support & Training- Headed by IS Manager: User Support

c) Each department has 4+ roles with clear reporting lines to their IS Manager, who reports to CIO.

d) Responsibilities of key posts

CIO: Set IT strategy, manage IT budget, ensure compliance with banking laws, report to CEO/Board.

IS Manager: Systems Development: Manage dev projects, ensure delivery on time, allocate analysts and programmers.

Senior Systems Analyst: Gather bank requirements for new products like loans, write SRS documents.

Systems Designer: Design core banking system modules and database schemas.

Senior Programmer: Develop and maintain core banking and mobile app code.

QA Tester: Test systems before release to avoid financial errors.

IS Manager: Infrastructure: Ensure 24/7 network uptime for branches and ATMs.

Network Administrator: Configure secure VPNs between branches.

IS Manager: Database & Security: Ensure customer data is encrypted and backed up.

DBA: Manage Oracle/SQL databases for millions of transactions.

Information Security Officer: Prevent fraud, manage access control, POPIA compliance.

IS Manager: User Support: Ensure staff get IT help quickly.

Help Desk Agent: Log and resolve staff tickets.

e) Motivation for Organizational Chart

1. Size of Bank: Large bank with many branches needs specialized teams. Separate units prevent bottlenecks.

2. Security & Compliance: Centralized structure with a dedicated Database & Security unit ensures POPIA, FICA and SARB compliance. Critical for a bank.

3. Core vs Support Systems: Splitting Development from Infrastructure ensures core banking systems are stable while new features are built.

4. 24/7 Operations: Multiple staff in Infrastructure and Support ensure ATMs, online banking and branch systems have no downtime.

5. Specialized Systems: banks use core banking, fraud detection, and mobile apps. Having separate Development and Security teams allows expertise in each.